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# Eval-Pair Matrix: Answer-Paired Meta-Evaluation of LLM Judges for Grounded RAG

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**Submission Track:** Track 1: Direct Submission  
**Submission Type:** Archival Long Paper (8 pages limitation for the main body)  
**Voluntary Reviewers:** Sriram Selvam (/profile?id=~Sriram\_Selvam1), Anneswa Ghosh (/profile?id=~Anneswa\_Ghosh1)  
**TL;DR:** We introduce Eval-Pair Matrix to test same-model bias in RAG evaluation, find little evidence of broad missed-error bias, and recommend full matrices, answer pairing, behavior strata, and label-task checks.  
**Abstract:** LLM-as-a-judge evaluation is widely used for retrieval-augmented generation (RAG), but reusing the same model family as both generator and judge makes self-leniency difficult to identify. We introduce Eval-Pair Matrix, a controlled meta-evaluation protocol for source-grounded RAG. Starting from GaRAGE questions and grounding passages, we induce one hidden answer-causal contradiction per record, generate answers from perturbed passages with GPT, Grok, and Gemini models, and then use the same models as blind judges to evaluate each answer against the original passages. The experiment contains 300 core records, 897 labeled generator outputs, and 2,683 judge verdicts in a crossed 3×3 matrix; the primary analysis uses 275 fully validated records. Instead of comparing diagonal and off-diagonal cells across different answers, we estimate same-model effects by pairing judges on the exact same candidate answer. This changes the interpretation: diagonal and off-diagonal F1 are similar, and the paired same-model recall effect is near zero (−0.5 pp; 95% cluster-bootstrap CI [−2.7, +1.7]). The only robust paired gap is lower matching-judge flagging for answers that avoided the induced claim (−4.3 pp). A targeted human evaluation finds that reviewed apparent false positives are alternate source-error detections, mistakes in labeling whether the induced claim was adopted, or unclear cases; none were adjudicated as genuine false alarms. The lesson is methodological: RAG judge studies should report full matrices, answer-paired effects, behavior strata, and label-task alignment.  
**Keywords:** LLM-as-a-judge, retrieval-augmented generation, RAG evaluation, meta-evaluation, same-model bias, self-leniency, source grounding, factuality evaluation, answer-paired evaluation  
**Confirmation:** I have read and agree with the workshop's policy on behalf of myself and my co-authors.  
**Submission Number:** 68

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## Paper Decision

Decision by Program Chairs 📅 29 Aug 2026, 21:19 (modified: 29 Aug 2026, 21:56) 👁 Program Chairs, Authors  
 📄 Revisions (/revisions?id=gHpSVb9rVW)

**Decision:** Accept

## Meta Review of Submission68 by Area Chair kgjx

Meta Review by Area Chair kgjx 📅 05 Aug 2026, 17:08 (modified: 31 Aug 2026, 19:41)  
 👁 Area Chairs, Authors, Program Chairs 📄 Revisions (/revisions?id=WxRwYuB4o1)

### Metareview:

The paper presents a thoughtful methodological framework for evaluating same-model bias in grounded RAG judging. I find the answer-paired design particularly valuable, as it compares matching and non-matching judges on the exact same candidate answer and therefore avoids several confounds present in raw diagonal-versus-off-diagonal comparisons. The crossed evaluation, behavior stratification, bootstrap analysis, and targeted human audit further strengthen the study.

The main limitations are that only one model endpoint is evaluated per provider family, making model identity difficult to separate from provider/style effects, and that the adoption labels rely on a single LLM evaluator with relatively limited human validation. These limitations constrain the generality of the null same-model effect but do not undermine the methodological contribution.

Overall, I find the evaluation protocol useful and relevant to the workshop and recommend acceptance.

**Recommendation:** Accept (Poster)

**Confidence:** 4: The area chair is confident but not absolutely certain

## Strong Methodological Contribution for Evaluating Same-Model Bias in RAG Judges

Official Review by Reviewer QVWs 📅 13 Jul 2026, 18:40 (modified: 31 Aug 2026, 19:45)  
 👁 Program Chairs, Area Chairs, Reviewer QVWs, Authors 📄 Revisions (/revisions?id=lqGyEBCZSE)

### Review:

This paper tackles a really common blind spot in RAG evaluation: same-model bias (or self-leniency) when the exact same LLM family acts as both the text generator and the judge. The introduction of the Eval-Pair Matrix is a fantastic methodological step forward. By isolating hidden contradictions and crossing outputs from GPT, Grok, and Gemini, the authors set up a highly controlled and convincing experiment. Finding that the same-model recall effect is actually near zero goes against some common industry assumptions, making this a genuinely valuable reality check for the field.

The main weakness here is that the experiment relies entirely on three massive, proprietary model families, which makes it hard to know if these findings apply to smaller, open-source models that might have completely different bias quirks. To improve the manuscript, it would be great if the authors briefly discussed how their evaluation matrix might scale or perform when testing smaller, locally hosted LLMs. Additionally, adding a short note on the practical computational cost of running these full, crossed 3x3 matrices would help practitioners figure out if this level of rigor is actually feasible for their everyday workflows.

My decision is to accept this paper because the Eval-Pair Matrix provides a highly controlled and much-needed approach to testing same-model bias in RAG judges. Even though the experiment only tests large proprietary models, the surprising findings challenge common industry assumptions and offer a valuable reality check for the field.

**Rating:** 8: Top 50% of accepted papers, clear accept

**Confidence:** 4: The reviewer is confident but not absolutely certain that the evaluation is correct

## Careful and honest meta-evaluation, but the headline is a null result on a small single-endpoint sample with a hard confound the design cannot resolve

Official Review by Reviewer LkgT  13 Jul 2026, 13:48 (modified: 31 Aug 2026, 19:45)

 Program Chairs, Area Chairs, Reviewer LkgT, Authors  Revisions (/revisions?id=wdATOfxISl)

### Review:

Summary. The paper proposes Eval-Pair Matrix, a controlled protocol for testing same-model self-leniency in source-grounded RAG judging. Starting from GaRAGe, it induces one hidden answer-causal contradiction per record, generates answers from perturbed passages with GPT/Grok/Gemini, and has the same three models judge each answer against the original grounding in a crossed 3×3 matrix. The central move is an answer-paired estimand: on the exact same candidate answer, does the matching judge behave differently from the two non-matching judges? The main finding is a near-zero paired recall effect (−0.5 pp, CI [−2.7, +1.7]) and a robust but reinterpreted avoided-claim flagging gap (−4.3 pp), which a targeted human evaluation attributes to label/task mismatch rather than self-leniency. Strengths.

The methodological argument is genuinely good. Reframing the raw diagonal-vs-off-diagonal contrast (which confounds self-leniency with generator difficulty, style, and refusal rates) into an answer-paired same-vs-cross contrast is the right identification move, and it is clearly explained. The pipeline is careful: five validation gates, leakage scans, behavior stratification, a two-layer label/judge separation, and cluster-bootstrap CIs by record ID. The human evaluation of apparent false positives is thoughtful and the SEC/FASB example is illuminating. The paper is admirably honest about interpretation — it explicitly refuses to read the avoided-claim gap as a lower false-alarm rate, and the limitations section is candid.

### Weaknesses.

The core finding is a null result, and the design cannot cleanly support it. By the authors' own admission, they evaluate exactly one endpoint per provider family, so the design "cannot separate model identity from provider-family or style-familiarity effects." This is fatal to the headline claim: "no robust same-model self-leniency" is being asserted from a setup that structurally cannot isolate the same-model variable. A near-zero effect under an unidentified confound is not evidence of absence. Sample size and single-annotator human eval. 275 validated records, and the crucial FP reinterpretation rests on 25 reviewed apparent false positives adjudicated by a single adjudicator with no double-annotation or agreement statistics. The entire "not self-leniency, but label/task mismatch" narrative — the paper's main interpretive contribution — hangs on 25 single-rater judgments. The rule-of-three upper bound (~12%) is acknowledged, but that is a wide margin for the central claim. The labels come from one GPT-5.4 label-evaluator. Both behavior labels and induced-error adoption labels are generated by a single model that is also one of the judged families. This introduces exactly the kind of circularity the paper is trying to study, one level down, and the human eval itself found ~8% adoption-label issues in reviewed refusal/conflict cases. The measurement instrument is entangled with the object of measurement. The one "robust" effect is reinterpreted into a null. The avoided-claim gap (−4.3 pp) is the only statistically robust paired result, and the paper spends considerable effort arguing it is not self-leniency but a label/task artifact. That argument is reasonable, but it means the paper's empirical yield is: one null (recall) and one robust effect explained away. The remaining takeaways are methodological recommendations (report full matrices, paired effects, behavior strata) that, while sound, are somewhat expected best practice. Per-generator paired effects are inconsistent and uncorrected (e.g., Gemini shows +6.0 pp recall Δ, GPT/Grok negative), which the paper flags as exploratory — but this heterogeneity is itself evidence that a single-endpoint design is picking up model-specific quirks rather than a general same-model phenomenon, undercutting the generality of the null.

Minor. Model version strings (GPT-5.4, Grok 4.3, Gemini 3.5 Flash) and a 2026 reference (REFLECT) should be double-checked for consistency. The distinction between "same-model" and "provider-family" is stated but the abstract and conclusion still read as broader claims than the single-endpoint design licenses. Recommendation. This is the most methodologically thoughtful of the papers I reviewed, and I would be glad to see the protocol developed. But as submitted, the headline is a null result produced by a design that admittedly cannot identify the variable of interest, resting on a small validated set and a 25-case single-adjudicator human study, with the one robust effect

reinterpreted away. The methodological framing is valuable but the empirical contribution is too thin and too confounded to accept. Reject, with strong encouragement to add multiple endpoints per family, independent/human labeling, and a double-annotated FP audit.

**Rating:** 4: Ok but not good enough - rejection  
**Confidence:** 4: The reviewer is confident but not absolutely certain that the evaluation is correct

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